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A NEW DERIVATION OF THE GENERAL EQUATIONS OF ELASTIC SHELLS*

P. M. NAGHDI
University of California, Berkeley, California, U.S.A.

Abstract—This paper is concerned with a new derivation of the general equations of the linear theory of elastic shells under the Kirchhoff–Love hypothesis. The entire boundary-value problem of shell theory is recast in terms of new variables for the strain measures as well as the stress and couple resultants. Particular attention is paid to an exact derivation of the constitutive equations and their first approximations which meet all invariance requirements. The natural boundary conditions for stress and couple resultants and all field equations consisting of compatibility, equilibrium and the constitutive equations (or their first approximations) involve only symmetric tensors and are, moreover, remarkably free of the anti-symmetric parts of both the middle surface strains and the couple resultants.

1. INTRODUCTION

MANY of the existing approximate constitutive equations in the linear bending theory of thin elastic shells, e.g. the various versions of Love's first approximation, are unsatisfactory in that they violate one or more of the invariance requirements.† Furthermore, most of the derivations of shell theory employ assumptions which are either unsatisfactory or not necessary. The desire for a rigorous development of the linear theory of elastic shells and the search for a satisfactory first approximation, especially with reference to the constitutive relations, has had a revival of interest during the past few years. Inasmuch as the nature of the equilibrium equations (in terms of stress and couple resultants) and the boundary conditions are well understood, the main issues are concerned with the character of deformation and suitable choice of strain measures for shells, a systematic derivation of the constitutive equations and an appraisal of their accuracy as compared with the three-dimensional equations.

The efforts directed in the recent past toward the resolution of these issues may be roughly divided into two categories: On the one hand, within the confine of elastic shell theory, several investigators have inferred or postulated the existence of a strain energy function for shells (per unit area of the middle surface) and, by an order of magnitude consideration and/or energy and related arguments, have endeavoured to obtain a first approximation theory. Others, beginning with the three-dimensional field equations of linear elasticity and by utilizing suitable scaling for displacements and stresses together with orders of magnitude consideration of certain parameters, have sought a systematic derivation of an appropriate system of differential equations and their solutions by successive approximations. Among the former, mention should be made of the contributions of Novozhilov (see [2; Ch. 1]),‡ Sanders [3], Koiter [4, 5], and Naghdi [1; Sec. 6]. In the latter

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† For an account of these invariance requirements in shell theory, see [1; Sec. 6] where a number of existing approximate constitutive relations is discussed.

‡ As noted by Novozhilov [2; p. 54], his approximate constitutive equations first derived in 1944 were also obtained independently and almost simultaneously by L. I. Balukh. Unfortunately Novozhilov's constitutive relations do not meet the requirement of co-ordinate invariance [1; Sec. 6].

category, we cite the papers of Johnson and Reissner [6] and Reiss [7]* for cylindrical shells and the more general recent investigation of Green [9].

The present paper aims to supply under the Kirchhoff–Love hypothesis† (1) an exact and systematic derivation of the linear theory of elastic shells, in particular its constitutive equations, and (2) a satisfactory first approximation theory. To achieve these goals, the entire boundary-value problem of elastic shell theory is recast in terms of new variables. A number of ingredients of the present derivation, either directly or by inference, have their counterparts in a number of earlier contributions, especially those of Novozhilov [2], Sanders [3] and Koiter [4].

Specifically the content of the paper is as follows: in Sections 2 and 3 following some preliminary results, new strain measures and stress and couple resultants are introduced with *justification* and are used to rewrite the equations of compatibility and equilibrium. A version of the static-geometric analogy, prominent in the Russian Literature, is also discussed briefly in terms of the new variables in Section 3. Section 4 contains an exact derivation of the constitutive equations and the natural boundary conditions under the Kirchhoff–Love hypothesis, and is followed in Section 5 by explicit development of the constitutive equations (valid for anisotropic shells) together with a satisfactory first approximation to the constitutive relations which meet all invariance requirements. Finally, in Section 6 comparison is made with earlier contributions, especially the recent result of Koiter [4].

Throughout the paper, unnecessary assumptions and groundless definitions are avoided, and it is particularly noteworthy that all field equations, including the approximate constitutive relations, as well as the derived boundary conditions for stress and couple resultants, exhibit the remarkable feature of being free of the anti-symmetric parts of both the middle surface strains and the couple resultants. It should also be noted that the foregoing field equations (including the approximate constitutive relations) admit all known general theorems for shells under the Kirchhoff–Love hypothesis, e.g. uniqueness, variational theorems, and the analogue of Betti's reciprocity theorem. As noted previously [1; Sec. 7], however, the existence theorem so far does not appear to have received attention.

The notation employed is essentially that of reference [1] where the *shell space* is referred to a Euclidean space of normal co-ordinates. Greek indices will have the range 1, 2, while Latin indices will assume the range 1, 2, 3. Also as in [1], double strokes are used to designate covariant differentiation with respect to the first fundamental form of the middle surface.

2. PRELIMINARIES

We recall that under the Kirchhoff–Love hypothesis, the shifted displacements have the form

$$\bar{u}_\alpha = v_\alpha(x^\beta) + x^3\beta_\alpha(x^\beta), \quad u_3 = w(x^\beta) \tag{2.1}‡$$

* The method of approach utilized in the papers of Johnson and Reissner [6] and Reiss [7] is analogous to that used earlier by Goodier [8] in the case of flat plates.

† Recall that according to this hypothesis normals to the undeformed middle surface of the shell remain normal to the deformed middle surface and suffer no extension.

‡ As pointed out previously [1], within the scope of the linear theory, the functional form of (2.1) is exact under the Kirchhoff–Love hypothesis.

and the nonvanishing strain measures are given by

$$2e_{\alpha\beta} = \mu^{\gamma}_{\alpha}(\gamma_{\gamma\beta}+x^3\kappa_{\gamma\beta})+\mu^{\gamma}_{\beta}(\gamma_{\gamma\alpha}+x^3\kappa_{\gamma\alpha}), \tag{2.2}$$

where

$$\gamma_{\alpha\beta} = v_{\alpha\|\beta}-b_{\alpha\beta}w, \quad \kappa_{\alpha\beta} = \beta_{\alpha\|\beta}, \quad \beta_{\alpha} = -(b^{\beta}_{\alpha}v_{\beta}+w_{\|\alpha}). \tag{2.3}$$

The compatibility equations associated with the strain measures (2.3) are

$$\bar{\varepsilon}^{\beta\gamma}(\bar{\varepsilon}^{\theta\alpha}\kappa_{\alpha\beta\|\gamma}+b^{\theta}_{\lambda}\bar{\varepsilon}^{\lambda\alpha}\gamma_{\alpha\beta\|\gamma}) = 0, \quad \bar{\varepsilon}^{\alpha\delta}\bar{\varepsilon}^{\beta\gamma}(\gamma_{\alpha\beta\|\gamma\delta}-\kappa_{\beta\alpha}b_{\delta\gamma}) = 0, \quad \bar{\varepsilon}^{\alpha\beta}(\kappa_{\alpha\beta}+b^{\nu}_{\alpha}\gamma_{\nu\beta}) = 0. \tag{2.4*}$$

Similarly, we recall the definition of stress and couple resultants in the form

$$\begin{Bmatrix} N^{\alpha\beta} \\ M^{\alpha\beta} \end{Bmatrix} = \int_{-h/2}^{+h/2} \mu\sigma^{\alpha\gamma}\mu^{\beta}_{\gamma}\begin{Bmatrix} 1 \\ x^3 \end{Bmatrix}\mathrm{d}x^3, \quad Q^{\alpha} = \int_{-h/2}^{+h/2}\mu\sigma^{\alpha 3}\mathrm{d}x^3 \tag{2.5}$$

and further recall the equations of equilibrium

$$\begin{aligned} &N^{\beta\alpha}{}_{\|\beta}-b^{\alpha}_{\gamma}Q^{\gamma}+p^{\alpha} = 0, \quad Q^{\alpha}{}_{\|\alpha}+b_{\alpha\gamma}N^{\gamma\alpha}+p = 0, \\ &M^{\beta\alpha}{}_{\|\beta}-Q^{\alpha}+m^{\alpha} = 0, \quad \bar{\varepsilon}_{\alpha\beta}(N^{\beta\alpha}-b^{\beta}_{\gamma}M^{\gamma\alpha}) = 0 \end{aligned} \tag{2.6}$$

which after elimination of the shear stress resultant Q^{α} from the first three of (2.6) assume the form

$$\begin{aligned} &N^{\beta\alpha}{}_{\|\beta}-b^{\alpha}_{\gamma}M^{\beta\gamma}{}_{\|\beta}-b^{\alpha}_{\gamma}m^{\gamma}+p^{\alpha} = 0, \\ &M^{\beta\alpha}{}_{\|\beta\alpha}+b_{\alpha\beta}N^{\alpha\beta}+p+m^{\alpha}{}_{\|\alpha} = 0, \\ &\bar{\varepsilon}_{\alpha\beta}(N^{\beta\alpha}-b^{\beta}_{\gamma}M^{\gamma\alpha}) = 0. \end{aligned} \tag{2.7}$$

In the foregoing equations, σ^{ij} is the symmetric stress tensor, $\{p^{\alpha}, p\}$ and $\{m^{\alpha}, 0\}$ represent the resultants due to effective external force and couple vectors measured per unit area of the middle surface, h is the thickness of the shell, $\bar{\varepsilon}_{\alpha\beta}$ and $\bar{\varepsilon}^{\alpha\beta}$ are the two-dimensional ε-system for the middle surface, $a_{\alpha\beta}$ and $b_{\alpha\beta}$ are, respectively, the first and second fundamental forms,

$$\mu^{\alpha}_{\beta} = (\delta^{\alpha}_{\beta}-x^3b^{\alpha}_{\beta}), \quad \mu = |\mu^{\alpha}_{\beta}| = \left(\frac{g}{a}\right)^{\frac{1}{2}} \tag{2.8}$$

and a and g are the determinants of the middle surface and the space metric, respectively.

3. MODIFIED EQUATIONS OF COMPATIBILITY AND EQUILIBRIUM. STATIC-GEOMETRIC ANALOGY

We now proceed to obtain with *justification* another representation for the compatibility and equilibrium equations. To this end, first define $\tilde{\kappa}_{\alpha\beta}$ through

$$\begin{aligned} \tilde{\kappa}_{\alpha\beta} &= \kappa_{\alpha\beta}-b^{\lambda}_{\beta}\gamma_{\lambda\alpha} \\ &= \kappa_{(\alpha\beta)}-b^{\lambda}_{(\beta}\gamma_{\lambda\alpha)} \end{aligned} \tag{3.1†}$$

* Equations of compatibility equivalent to (2.4) were first derived by Gol'denveizer, [10]. For another derivation leading to (2.4), see [1; Sec. 4].

† That the anti-symmetric part of $\tilde{\kappa}_{\alpha\beta}$, namely $\tilde{\kappa}_{[\alpha\beta]}$ vanishes identically follows from $(2.4)_3$. Hence $\tilde{\kappa}_{\alpha\beta} = \tilde{\kappa}_{(\alpha\beta)}$.

the symmetry of which is ensured by $(2.4)_3$ and decompose the second order tensor $\gamma_{\alpha\beta}$ into symmetric and anti-symmetric parts, i.e.

$$\gamma_{\alpha\beta} = \gamma_{(\alpha\beta)} + \gamma_{[\alpha\beta]} \tag{3.2}$$

so that with the aid of $(2.3)_1$

$$\begin{aligned} \gamma_{(\alpha\beta)} &= \tfrac{1}{2}(\gamma_{\alpha\beta} + \gamma_{\beta\alpha}) = \tfrac{1}{2}(v_{\alpha\|\beta} + v_{\beta\|\alpha}) - b_{\alpha\beta} w\,, \\ \gamma_{[\alpha\beta]} &= \tfrac{1}{2}(\gamma_{\alpha\beta} - \gamma_{\beta\alpha}) = \tfrac{1}{2}(v_{\alpha\|\beta} - v_{\beta\|\alpha})\,. \end{aligned} \tag{3.3}$$

It should be evident that whereas $\gamma_{(\alpha\beta)}$ is the unique symmetric part of $\gamma_{\alpha\beta}$, $\tilde{\kappa}_{\alpha\beta} = \tilde{\kappa}_{\beta\alpha}$ is not the unique symmetric part of $\kappa_{\alpha\beta}$.

Introducing (3.1) and (3.2) into the first two of (2.4) and taking advantage of the symmetry and anti-symmetry of the various terms, we obtain the following modified equations of compatibility

$$\begin{aligned} \bar{\varepsilon}^{\beta\gamma}[\bar{\varepsilon}^{\theta\alpha}(\tilde{\kappa}_{\alpha\beta\|\gamma} + b^{\lambda}_{\beta}\gamma_{(\lambda\alpha)\|\gamma}) + b^{\theta}_{\lambda}\bar{\varepsilon}^{\lambda\alpha}\gamma_{(\alpha\beta)\|\gamma}] &= 0\,, \\ \bar{\varepsilon}^{\alpha\delta}\bar{\varepsilon}^{\beta\gamma}[\gamma_{(\alpha\beta)\|\gamma\delta} - b_{\delta\gamma}\tilde{\kappa}_{\beta\alpha} - b_{\gamma\delta}b^{\lambda}_{\beta}\gamma_{(\lambda\alpha)}] &= 0\,. \end{aligned} \tag{3.4}$$

It is remarkable that equation (3.4) are free from the anti-symmetric $\gamma_{[\alpha\beta]}$.

We now return to the equilibrium equations (2.7), define (analogously to $\tilde{\kappa}_{\alpha\beta}$) a tensor $\tilde{N}^{\alpha\beta}$ through

$$\tilde{N}^{\alpha\beta} = N^{\alpha\beta} + b^{\beta}_{\lambda} M^{\lambda\alpha} \tag{3.5}$$

the symmetry of which is ensured by $(2.7)_3$ and also write

$$M^{\alpha\beta} = M^{(\alpha\beta)} + M^{[\alpha\beta]}\,, \tag{3.6}$$

where

$$M^{(\alpha\beta)} = \tfrac{1}{2}(M^{\alpha\beta} + M^{\beta\alpha})\,, \quad M^{[\alpha\beta]} = \tfrac{1}{2}(M^{\alpha\beta} - M^{\beta\alpha})\,. \tag{3.7}$$

Substituting (3.5) and (3.6) into the first two of (2.7), taking advantage of the anti-symmetry of $M^{[\alpha\beta]}$, and observing that

$$M^{[\beta\alpha]}{}_{\|\beta\alpha} = b_{\alpha\beta} b^{\beta}_{\gamma} M^{[\gamma\alpha]} \equiv 0\,,$$

we deduce

$$\begin{aligned} \tilde{N}^{\alpha\beta}{}_{\|\beta} - b^{\alpha}_{\beta\|\gamma} M^{(\beta\gamma)} - 2b^{\alpha}_{\beta} M^{(\beta\gamma)}{}_{\|\gamma} + r^{\alpha} &= 0\,, \\ b_{\alpha\beta}\tilde{N}^{\alpha\beta} - b_{\alpha\beta} b^{\beta}_{\gamma} M^{(\alpha\gamma)} + M^{(\alpha\beta)}{}_{\|\alpha\beta} + r &= 0\,, \end{aligned} \tag{3.8}$$

where

$$r^{\alpha} = p^{\alpha} - b^{\alpha}_{\beta} m^{\beta}\,, \quad r = p + m^{\alpha}{}_{\|\alpha}\,. \tag{3.9}$$

The modified equations of equilibrium (3.8) involve only $\tilde{N}^{\alpha\beta}$ and $M^{(\alpha\beta)}$ and are remarkably free of the anti-symmetric $M^{[\alpha\beta]}$.

In view of the structure of the compatibility and equilibrium equations (3.4) and (3.8), it is natural to inquire if it is possible to express the original compatibility and equilibrium equations (2.4) and (2.7) in terms of symmetric tensors alone when both strain measures $\gamma_{\alpha\beta}$ and $\kappa_{\alpha\beta}$ and both $N^{\alpha\beta}$ and $M^{\alpha\beta}$ are decomposed uniquely into their respective symmetric and anti-symmetric parts. For definiteness, consider the case of compatibility equations and as in (3.2), also write $\kappa_{\alpha\beta}$ in the form

$$\kappa_{\alpha\beta} = \kappa_{(\alpha\beta)} + \kappa_{[\alpha\beta]}\,. \tag{3.10}$$

Then, with the aid of

$$\gamma_{[\alpha\beta]} = \gamma'\bar{\varepsilon}_{\alpha\beta}, \quad \kappa_{[\alpha\beta]} = \kappa'\bar{\varepsilon}_{\alpha\beta} = \tfrac{1}{2}(b^{\lambda}_{\beta}\gamma_{\lambda\alpha} - b^{\lambda}_{\alpha}\gamma_{\lambda\beta}), \tag{3.11}$$

where the second of $(3.11)_2$ follows from $(2.4)_3$ and γ' and κ' are scalars, it can easily be shown that substitution of (3.2) and (3.10) into $(2.4)_2$ will result in an equation free from $\gamma_{[\alpha\beta]}$ and $\kappa_{[\alpha\beta]}$, but $(2.4)_1$ will contain both κ' and γ'. However, in the latter equation, κ' can be eliminated with the use of the second of $(3.11)_2$ so that the resulting equations read

$$\begin{gathered}\bar{\varepsilon}^{\beta\gamma}[\bar{\varepsilon}^{\theta\alpha}\kappa_{(\alpha\beta)\|\gamma} + b^{\theta}_{\lambda}\bar{\varepsilon}^{\lambda\alpha}\gamma_{(\alpha\beta)\|\gamma}] - \tfrac{1}{2}\bar{\varepsilon}^{\theta\gamma}[\bar{\varepsilon}^{\alpha\beta}b^{\lambda}_{\beta}\gamma_{(\lambda\alpha)}]_{\|\gamma} = b^{\theta}_{\lambda}\bar{\varepsilon}^{\lambda\gamma}\gamma'_{\|\gamma} - \tfrac{1}{2}\bar{\varepsilon}^{\theta\gamma}(b^{\lambda}_{\lambda}\gamma')_{\|\gamma}, \\ \bar{\varepsilon}^{\alpha\delta}\bar{\varepsilon}^{\beta\gamma}[\gamma_{(\alpha\beta)\|\gamma\delta} - \kappa_{(\beta\alpha)}b_{\delta\gamma}] = 0.\end{gathered} \tag{3.12}$$

Whereas the compatibility equations (3.4) involve $\gamma_{(\alpha\beta)}$ and $\tilde{\kappa}_{\alpha\beta}$, equations (3.12) contain $\gamma_{(\alpha\beta)}$ and $\kappa_{(\alpha\beta)}$, as well as γ'. The further elimination of γ' (due to the anti-symmetric part of $\gamma_{\alpha\beta}$) from $(3.12)_1$ does not appear to be possible and this, in part, is the motivation for defining $\tilde{\kappa}_{\alpha\beta}$ in the form (3.1). It is clear that a parallel argument applies to the equilibrium equations if, instead of (3.5) and (3.6), we use (3.6) and decompose $N_{\alpha\beta}$ into its unique symmetric and anti-symmetric parts. The idea of reducing the number of stress and couple resultants on the one hand, and the number of independent strain measures on the other, apparently was suggested by Lur'e [11]* and is prominent in the works of Novozhilov [2], Sanders [3] and Koiter [4]. Further reference to these and their comparisons with the foregoing derivation, however, will be postponed until Section 6.

Before closing this section, we recall a significant analogy or correspondence theorem, introduced simultaneously by Lur'e [13] and Gol'denveizer [10] and rendered more complete by a recent paper of Lur'e [14]†. According to this analogy, referred to as the static-geometric analogy in the Russian literature, in the absence of external force $\{p^{\alpha}, p\}$ and couple $\{m^{\alpha}, 0\}$, a one-to-one correspondence exists between the equations of equilibrium and the compatibility equations. The truth of this assertion in terms of the new variables $\tilde{N}^{\alpha\beta}$, $M^{(\alpha\beta)}$, $\gamma^{(\alpha\beta)}$ and $\tilde{\kappa}_{\alpha\beta}$ is easily verified if we observe the following correspondence

$$\tilde{N}^{\alpha\beta} \leftrightarrow -\bar{\varepsilon}^{\alpha\mu}\bar{\varepsilon}^{\beta\nu}[\tilde{\kappa}_{\mu\nu} + b^{\lambda}_{\nu}\gamma_{(\lambda\mu)}] + b^{\alpha}_{\lambda}\bar{\varepsilon}^{\lambda\mu}\bar{\varepsilon}^{\beta\nu}\gamma_{(\mu\nu)}, \quad M^{(\alpha\beta)} \leftrightarrow \bar{\varepsilon}^{\alpha\mu}\bar{\varepsilon}^{\beta\nu}\gamma_{(\mu\nu)} \tag{3.13}$$

substitution of which into (3.8) with $p^{\alpha} = p = m^{\alpha} = 0$ yields the compatibility equations (3.4).

Now let

$$P^{\alpha\beta} = \tilde{N}^{\alpha\beta} - ik\bar{\varepsilon}^{\beta\nu}\{\bar{\varepsilon}^{\alpha\mu}[\tilde{\kappa}_{\mu\nu} + b^{\lambda}_{\nu}\gamma_{(\lambda\mu)}] + b^{\alpha}_{\lambda}\bar{\varepsilon}^{\lambda\mu}\gamma_{(\mu\nu)}\}, \quad Q^{\alpha\beta} = M^{(\alpha\beta)} + ik[\bar{\varepsilon}^{\alpha\mu}\bar{\varepsilon}^{\beta\nu}\gamma_{(\mu\nu)}], \tag{3.14}$$

where $i = (-1)^{\frac{1}{2}}$ and k is an arbitrary scalar. Then with the aid of (3.14) and by suitable combination of (3.4) and (3.8), we obtain

$$P^{\alpha\beta}{}_{\|\beta} - b^{\alpha}_{\beta\|\gamma}Q^{\beta\gamma} - 2b^{\alpha}_{\beta}Q^{\beta\gamma}{}_{\|\gamma} + r^{\alpha} = 0 \tag{3.15}$$

$$b_{\alpha\beta}P^{\alpha\beta} - b_{\alpha\beta}b^{\beta}_{\gamma}Q^{\alpha\gamma} + Q^{\alpha\beta}{}_{\|\alpha\beta} + r = 0 \tag{3.16}$$

the real and imaginary parts of which yield, respectively, the equilibrium and compatibility equations (3.8) and (3.4).

* In this connection, see also [12; Part I, Sec. 26].

† An account of earlier results of Lur'e [13] and Gol'denveizer [1] may also be found in the books by Novozhilov [2] and Gol'denveizer [12]. For a more general development of the correspondence theorem based on the works of Gol'denveizer and Lur'e, see [1; Sec. 7]. The analogy is also suggested by the independent work of Sanders [3].

4. DERIVATION OF CONSTITUTIVE EQUATIONS AND BOUNDARY CONDITIONS

We now proceed to derive the constitutive equations of the linear theory of elastic shells together with the natural boundary conditions in terms of the newly defined strain measures (3.1) and (3.2) and the newly defined resultants (3.1) and (3.6). The derivation is accomplished by means of a virtual work principle.

The virtual work principle in 3-space (as given, e.g. by Green and Zerna [15]), after introducing (2.2) and (2.5) and integrating across the thickness of the shell from $-h/2$ to $+h/2$, may be written as

$$\delta U = \delta A_p + \delta A_c, \tag{4.1}$$

where

$$\delta A_p = \int_s (p^\alpha \delta v_\alpha + m^\alpha \delta\beta_\alpha + p\delta w)\mathrm{d}a, \quad \delta A_c = \int_c \upsilon_\alpha (N^{*\alpha\beta}\delta v_\beta + M^{*\alpha\beta}\delta\beta_\beta + Q^{*\alpha}\delta w)\mathrm{d}s, \tag{4.2}$$

and s and c refer to the middle surface and its edge curve, respectively, and the starred quantities stand for the edge resultants. Consider first $(4.2)_1$ which after substitution from the first three of the equilibrium equations (2.6) reads

$$\delta A_p = \int_s [-(N^{\beta\alpha}{}_{\|\beta} - b^\alpha_\gamma Q^\gamma)\delta v_\alpha - (M^{\beta\alpha}{}_{\|\beta} - Q^\alpha)\delta\beta_\alpha - (Q^\alpha{}_{\|\alpha} + b_{\alpha\gamma}N^{\gamma\alpha})\delta w]\mathrm{d}a$$

and which, with the use of expressions of the type $(N^{\beta\alpha}\delta v_\alpha)_{\|\beta} = N^{\beta\alpha}{}_{\|\beta}\delta v_\alpha + N^{\beta\alpha}\delta v_{\alpha\|\beta}$, may also be written as

$$\begin{aligned} -\delta A_p = &\int_s [(N^{\beta\alpha}\delta v_\alpha)_{\|\beta} + (M^{\beta\alpha}\delta\beta_\alpha)_{\|\beta} + (Q^\alpha \delta w)_{\|\alpha}]\mathrm{d}a \\ &+ \int_s [-N^{\beta\alpha}(\delta v_{\alpha\|\beta} - b_{\beta\alpha}\delta w) - M^{\beta\alpha}\delta\beta_{\alpha\|\beta} - Q^\alpha(\delta w_{\|\alpha} + \delta\beta_\alpha + b^\beta_\alpha \delta v_\beta)]\mathrm{d}a. \end{aligned}$$

Observing that in the second integral the coefficient of $N^{\beta\alpha}$ is $\delta\gamma_{\alpha\beta}$ and that by virtue of $(2.3)_3$ the coefficient of Q^α vanishes, the foregoing expression after the application of the divergence theorem to the first integral becomes

$$\begin{aligned} \delta A_p = &\int_s (N^{\beta\alpha}\delta\gamma_{\alpha\beta} + M^{\beta\alpha}\delta\kappa_{\alpha\beta})\mathrm{d}a \\ &- \int_c \upsilon_\beta (N^{\beta\alpha}\delta v_\alpha + M^{\beta\alpha}\delta\beta_\alpha + Q^\beta \delta w)\mathrm{d}s. \end{aligned} \tag{4.3}$$

If in the line integral in (4.3) as well as $(4.2)_2$ the terms involving $\partial/\partial s(\delta w)$ are integrated by parts, and if we introduce the notation

$$G' = M^{\alpha\beta}\upsilon_\alpha \upsilon_\beta, \quad H' = \bar{\varepsilon}_{\beta\lambda} M^{\alpha\beta}\upsilon_\alpha \upsilon^\lambda, \tag{4.4}$$

then the substitution of (4.3) and $(4.2)_2$ into (4.1) results in

$$\delta U = \int_s (N^{\beta\alpha}\delta\gamma_{\alpha\beta} + M^{\beta\alpha}\delta\kappa_{\alpha\beta})\mathrm{d}a$$

$$+\int_c \left\{ v_\alpha[(N^{*\alpha\beta} - b^\beta_\lambda M^{*\alpha\lambda}) - (N^{\alpha\beta} - b^\beta_\lambda M^{\alpha\lambda})]\delta v_\beta \right.$$

$$\left. -(G'^* - G')\delta\left(\frac{\partial w}{\partial v}\right) + \left[\left(Q^{*\alpha}v_\alpha - \frac{\partial H'^*}{\partial s}\right) - \left(Q^\alpha v_\alpha - \frac{\partial H'}{\partial s}\right)\right]\delta w \right\}\mathrm{d}s$$

$$+\sum_{i=1}^{N} \{[H'^*(s_i+o) - H'^*(s_i-o)] - [H'(s_i+o) - H'(s_i-o)]\}\delta w(s_i), \tag{4.5}$$

where s_i, $(i = 1, 2, \ldots N)$, are points on c at which H' has jump discontinuities and $\partial/\partial s$ and $\partial/\partial v$ stand, respectively, for the partial derivatives along the tangent and normal to c.

Consider now the integrand of the surface integral in (4.5), namely

$$I = N^{\beta\alpha}\delta\gamma_{\alpha\beta} + M^{\beta\alpha}\delta\kappa_{\alpha\beta}$$

which after substituting from (3.1), (3.2), (3.5) and (3.6), since all terms involving $\gamma_{[\alpha\beta]}$ and $M^{[\alpha\beta]}$ cancel, reduces simply to

$$I = \tilde{N}^{\beta\alpha}\delta\gamma_{(\alpha\beta)} + M^{(\beta\alpha)}\delta\tilde{\kappa}_{\alpha\beta}. \tag{4.6}$$

Thus, we may write the surface integral in (4.5) as

$$\delta\int_s \bar{\Sigma}\mathrm{d}a = \int_s [\tilde{N}^{\beta\alpha}\delta\gamma_{(\alpha\beta)} + M^{(\beta\alpha)}\delta\tilde{\kappa}_{\alpha\beta}]\mathrm{d}a, \tag{4.7}$$

where

$$\bar{\Sigma} = \int_{-h/2}^{+h/2} \mu\Sigma(e_{ij})\mathrm{d}x^3 \tag{4.8}$$

and Σ denotes the strain energy in 3-space. If the strain energy $\bar{\Sigma}$ is an exact differential, then

$$\delta\bar{\Sigma} = \frac{\partial\bar{\Sigma}}{\partial\gamma_{(\alpha\beta)}}\delta\gamma_{(\alpha\beta)} + \frac{\partial\bar{\Sigma}}{\partial\tilde{\kappa}_{\alpha\beta}}\delta\tilde{\kappa}_{\alpha\beta}$$

and by (4.7) we obtain

$$\tilde{N}^{\beta\alpha} = \frac{\partial\bar{\Sigma}}{\partial\gamma_{(\alpha\beta)}}, \quad M^{(\beta\alpha)} = \frac{\partial\bar{\Sigma}}{\partial\tilde{\kappa}_{\alpha\beta}} \tag{4.9}$$

as the constitutive equations. If we admit the inverse of (4.9), then through a Legendre transformation

$$\bar{W} = [\tilde{N}^{\beta\alpha}\gamma_{(\alpha\beta)} + M^{(\beta\alpha)}\tilde{\kappa}_{\alpha\beta}] - \bar{\Sigma}, \tag{4.10}$$

where

$$\bar{W} = \int_{-h/2}^{+h/2} \mu W(\sigma^{ij})\mathrm{d}x^3, \tag{4.11}$$

$$W = \sigma^{ij}e_{ij} - \Sigma(e_{ij}),$$

the constitutive relations may be inverted in the form

$$\gamma_{(\alpha\beta)} = \frac{\partial \bar{W}}{\partial \tilde{N}^{\beta\alpha}}, \quad \tilde{\kappa}_{\alpha\beta} = \frac{\partial \bar{W}}{\partial M^{(\beta\alpha)}}. \tag{4.12}$$

Further developments of the constitutive relations will be carried out in the next section.

We now return to (4.5) and consider the line integral which yields the usual boundary conditions of the classical theory of shells under the Kirchhoff–Love hypothesis. However, since in the present derivation the compatibility equations (3.4), the equilibrium equations (3.8), and the constitutive equations (4.9), all involve symmetric tensors and are free from $\gamma_{[\alpha\beta]}$ and $M^{[\alpha\beta]}$, the boundary conditions must also be represented in terms of the new variables.

With reference to the line integral in (4.5), it is clear that the coefficient of δv_β with the aid of (3.5) and (3.6) becomes

$$\upsilon_\alpha(N^{\alpha\beta} - b^\beta_\lambda M^{\alpha\lambda}) = \upsilon_\alpha[\tilde{N}^{\alpha\beta} - 2b^\beta_\lambda M^{(\lambda\alpha)}]. \tag{4.13}$$

Similarly, since $M^{[\alpha\beta]}\upsilon_\alpha\upsilon_\beta = 0$, G' can be put in the form

$$G' = M^{(\alpha\beta)}\upsilon_\alpha\upsilon_\beta. \tag{4.14}$$

The coefficient of δw under the line integral, with the aid of $(2.6)_3$, $(4.4)_2$, and (3.6) can be written as

$$\begin{aligned} \upsilon_\alpha Q^\alpha - \frac{\partial H'}{\partial s} &= \upsilon_\alpha M^{(\beta\alpha)}{}_{\|\beta} - \frac{\partial}{\partial s}[\bar{\varepsilon}_{\beta\lambda} M^{(\alpha\beta)}\upsilon_\alpha\upsilon^\lambda] \\ &\quad + \upsilon_\alpha M^{[\beta\alpha]}{}_{\|\beta} - \frac{\partial}{\partial s}(\bar{\varepsilon}_{\beta\lambda} M^{[\alpha\beta]}\upsilon_\alpha\upsilon^\lambda). \end{aligned} \tag{4.15}$$

Further, since $M^{[\alpha\beta]}$ can be expressed in terms of a single scalar in the form

$$M^{[\alpha\beta]} = M'\bar{\varepsilon}^{\alpha\beta}, \tag{4.16}$$

then

$$\bar{\varepsilon}_{\beta\lambda} M^{[\alpha\beta]}\upsilon_\alpha\upsilon^\lambda = -\delta^\alpha_\lambda M'\upsilon_\alpha\upsilon^\lambda = -M'. \tag{4.17}$$

But the third term on the right-hand side of (4.15) may be written as

$$\upsilon_\alpha M^{[\beta\alpha]}{}_{\|\beta} = \upsilon_\alpha\bar{\varepsilon}^{\beta\alpha}M'{}_{\|\beta} = -M'{}_{\|\beta}\lambda^\beta = -\frac{\mathrm{d}M'}{\mathrm{d}s}, \tag{4.18}$$

where $\lambda^\beta = \mathrm{d}x^\beta/\mathrm{d}s = \upsilon_\alpha\bar{\varepsilon}^{\alpha\beta}$ is the unit tangent vector to c. By virtue of (4.17) and (4.18) the last two terms on the right-hand side of (4.15) cancel and we have

$$\upsilon_\alpha Q^\alpha - \frac{\partial H'}{\partial s} = \upsilon_\alpha M^{(\beta\alpha)}{}_{\|\beta} - \frac{\partial}{\partial s}[\bar{\varepsilon}_{\beta\lambda} M^{(\alpha\beta)}\upsilon_\alpha\upsilon^\lambda]. \tag{4.19}$$

Thus, in view of (4.13), (4.14), and (4.18), the boundary conditions are either

$$\begin{aligned} \upsilon_\alpha[\tilde{N}^{\alpha\beta} - 2b^\beta_\lambda M^{(\lambda\alpha)}] &= \upsilon_\alpha(N^{*\alpha\beta} - b^\beta_\lambda M^{*\alpha\lambda}) \\ M^{(\alpha\beta)}\upsilon_\alpha\upsilon_\beta &= M^{*\alpha\beta}\upsilon_\alpha\upsilon_\beta \\ \upsilon_\alpha M^{(\beta\alpha)}{}_{\|\beta} - \frac{\partial}{\partial s}[\bar{\varepsilon}_{\beta\lambda} M^{(\alpha\beta)}\upsilon_\alpha\upsilon^\lambda] &= \upsilon_\alpha Q^{*\alpha} - \frac{\partial}{\partial s}(\bar{\varepsilon}_{\beta\lambda} M^{*\alpha\beta}\upsilon_\alpha\upsilon^\lambda) \end{aligned} \tag{4.20}$$

or the displacements specified by v_β, w, and $\partial w/\partial v$.

5. EXPLICIT FORM OF THE CONSTITUTIVE EQUATIONS

If we assume the strain energy function Σ in (4.8) to be quadratic, then under the Kirchhoff–Love hypothesis

$$\begin{aligned}\Sigma &= \tfrac{1}{2}A^{\alpha\beta\gamma\delta}e_{\alpha\beta}e_{\gamma\delta}\\ &= \tfrac{1}{2}A^{\alpha\beta\gamma\delta}\mu^{\lambda}_{\alpha}\mu^{\rho}_{\gamma}[\gamma_{\lambda\beta}\gamma_{\rho\delta}+x^3(\gamma_{\lambda\beta}\kappa_{\rho\delta}+\kappa_{\lambda\beta}\gamma_{\rho\delta})+(x^3)^2\kappa_{\lambda\beta}\kappa_{\rho\delta}],\end{aligned} \tag{5.1*}$$

where (2.2) has been used. We now recall that the tensor coefficient $A^{\alpha\beta\lambda\delta}$ is related to the corresponding quantity on $x^3 = 0$ through (see [1; Sec. 3])

$$A^{\alpha\beta\gamma\delta}\mu^{\lambda}_{\alpha}\mu^{\rho}_{\gamma} = \bar{A}^{\lambda\theta\rho\nu}\overset{-1}{\mu}{}^{\beta}_{\theta}\overset{-1}{\mu}{}^{\delta}_{\nu} = \phi^{\lambda\beta\rho\delta}. \tag{5.2}$$

In (5.2), on account of symmetry of $A^{\alpha\beta\gamma\delta}$,

$$\phi^{\lambda\beta\rho\delta} = \phi^{\rho\delta\lambda\beta}, \tag{5.3}$$

and $\overset{-1}{\mu}{}^{\alpha}_{\beta}$ is the inverse of $(2.8)_1$ so that

$$\overset{-1}{\mu}{}^{\alpha}_{\lambda}\mu^{\lambda}_{\beta} = \overset{-1}{\mu}{}^{\alpha}_{\beta} - x^3\overset{-1}{\mu}{}^{\alpha}_{\lambda}b^{\lambda}_{\beta} \tag{5.4}$$

or

$$x^3\overset{-1}{\mu}{}^{\alpha}_{\lambda}b^{\lambda}_{\beta} = x^3 b^{\lambda}_{\beta}\overset{-1}{\mu}{}^{\alpha}_{\lambda} = \overset{-1}{\mu}{}^{\alpha}_{\beta} - \delta^{\alpha}_{\beta}. \tag{5.5}$$

Introducing (3.1), (3.2), and (5.2) into (5.1), using (5.5), observing various symmetries such as (5.3) and using $(2.8)_1$ whenever feasible, we obtain

$$\begin{aligned}\Sigma = \tfrac{1}{2}\bar{A}^{\lambda\theta\nu\rho}\overset{-1}{\mu}{}^{\beta}_{\theta}\overset{-1}{\mu}{}^{\delta}_{\nu}\{&[\delta^{\eta}_{\beta}\delta^{\varepsilon}_{\delta}+2x^3 b^{\eta}_{\beta}\delta^{\varepsilon}_{\delta}+(x^3)^2 b^{\eta}_{\beta}b^{\varepsilon}_{\delta}]\gamma_{(\eta\lambda)}\gamma_{(\varepsilon\rho)}\\ &+2x^3(\delta^{\eta}_{\beta}\delta^{\varepsilon}_{\delta}+x^3 b^{\eta}_{\beta}\delta^{\varepsilon}_{\delta})\gamma_{(\eta\lambda)}\tilde{\kappa}_{\varepsilon\rho}+(x^3)^2\tilde{\kappa}_{\lambda\beta}\tilde{\kappa}_{\rho\delta}\}\end{aligned} \tag{5.6}$$

Substitution of (5.6) into (4.8) followed by integration results in

$$\bar{\Sigma} = \tfrac{1}{2}[{}_0D^{\lambda\eta\rho\varepsilon}\gamma_{(\lambda\eta)}\gamma_{(\rho\varepsilon)}+2{}_1D^{\lambda\eta\rho\varepsilon}\gamma_{(\lambda\eta)}\tilde{\kappa}_{\rho\varepsilon}+{}_2D^{\lambda\eta\rho\varepsilon}\tilde{\kappa}_{\lambda\eta}\tilde{\kappa}_{\rho\varepsilon}], \tag{5.7}$$

where

$$\begin{aligned}&{}_nD^{\alpha\beta\rho\varepsilon} = {}_nD^{\beta\alpha\rho\varepsilon} = {}_nD^{\alpha\beta\varepsilon\rho} = {}_nD^{\rho\varepsilon\alpha\beta};\quad (n = 0, 1, 2),\\ &{}_0D^{\lambda\eta\rho\varepsilon} = {}_0B^{\lambda\eta\rho\varepsilon}+2b^{\eta}_{\beta}\,{}_1B^{\lambda\beta\rho\varepsilon}+b^{\eta}_{\beta}b^{\varepsilon}_{\delta}\,{}_2B^{\lambda\beta\rho\delta},\\ &{}_1D^{\lambda\eta\rho\varepsilon} = {}_1B^{\lambda\eta\rho\varepsilon}+b^{\eta}_{\beta}\,{}_2B^{\lambda\beta\rho\varepsilon},\\ &{}_2D^{\lambda\eta\rho\varepsilon} = \tfrac{1}{4}({}_2B^{\lambda\eta\rho\varepsilon}+{}_2B^{\rho\varepsilon\lambda\eta}+{}_2B^{\varepsilon\rho\eta\lambda}+{}_2B^{\rho\varepsilon\eta\lambda})\end{aligned} \tag{5.8}$$

and

$${}_nB^{\lambda\beta\rho\delta} = \int_{-h/2}^{+h/2}\mu\bar{A}^{\lambda\theta\rho\nu}\overset{-1}{\mu}{}^{\beta}_{\theta}\overset{-1}{\mu}{}^{\delta}_{\nu}(x^3)^n\mathrm{d}x^3;\quad (n = 0, 1, 2). \tag{5.9}$$

The linear constitutive equations sought now follow from (4.9) and (5.7) and read

$$\begin{aligned}\tilde{N}^{\alpha\beta} &= {}_0D^{\alpha\beta\rho\varepsilon}\gamma_{(\rho\varepsilon)}+{}_1D^{\alpha\beta\rho\varepsilon}\tilde{\kappa}_{\rho\varepsilon},\\ M^{(\alpha\beta)} &= {}_1D^{\alpha\beta\rho\varepsilon}\gamma_{(\rho\varepsilon)}+{}_2D^{\alpha\beta\rho\varepsilon}\tilde{\kappa}_{\rho\varepsilon}.\end{aligned} \tag{5.10}$$

* In obtaining (5.1) the stress component σ_{33} has been neglected and the coefficients $A^{\alpha\beta\gamma\delta}$ include the effect of e_{33}.

It is clear that the constitutive equations (5.10) meet the requirement of co-ordinate invariance, are in accord with the equations of equilibrium, especially $(2.7)_3$, and are also consistent with the virtual work principle from which, in fact, they were derived. In addition, it can be readily shown that they are form invariant under rigid displacement. The constitutive equations (5.10) also meet the reciprocity and all other general theorems including the variational theorems as discussed previously in [1].

To complete the explicit derivation of the constitutive equations, we now proceed to evaluate the coefficients ${}_nB^{\alpha\beta\gamma\delta}$ given by (5.9) which occur in (5.8). Since *$|x^3| \leq (h/2) < R$, where R stands for the least (local) radius of the curvature to the middle surface, the characteristic roots of $x^3 b^\alpha{}_\beta$ associated with $(2.8)_2$ are less than 1. Hence, according to a theorem of algebra (Mirsky [16; p. 332]), $\mu^{-1}{}^\alpha_\beta$ admits the convergent series representation

$$\mu^{-1}{}^\alpha_\beta = \sum_{n=0}^{\infty} \overset{n}{b}{}^\alpha_\beta (x^3)^n, \tag{5.11}$$

where

$$\overset{0}{b}{}^\alpha_\beta = \delta^\alpha_\beta, \quad \overset{1}{b}{}^\alpha_\beta = b^\alpha_\beta, \quad \overset{n}{b}{}^\alpha_\beta = b^\alpha_\lambda \overset{(n-1)}{b}{}^\lambda_\beta = b^\alpha_\rho b^\rho_\lambda \overset{(n-2)}{b}{}^\lambda_\beta = \ldots \tag{5.12}$$

If we restrict attention to elastic shells whose mechanical properties are homogenous in x^3 so that $\bar{A}^{\alpha\beta\gamma\delta}$ depend on surface co-ordinates alone and recall that

$$\mu\,\mu^{-1}{}^\alpha_\beta = \delta^\alpha_\beta + x^3(b^\alpha_\beta - 2H\delta^\alpha_\beta), \quad H = \tfrac{1}{2} b^\gamma_\gamma,$$

then after introducing (5.11) into (5.9) and carrying out the integration, ${}_nB^{\mu\beta\lambda\eta}$ may be expressed as

$$\begin{aligned} {}_nB^{\mu\beta\lambda\eta} = \bar{A}^{\mu\rho\lambda\delta} \Bigg\{ & \delta^\beta_\rho \delta^\eta_\delta \left(\frac{h}{2}\right)^{(n+1)} \frac{[1+(-1)^n]}{(1+n)} \\ & + \sum_{m=1}^{\infty} I^{\beta\nu}_{\eta\rho} \overset{(m-1)}{b}{}^\nu_\delta \left(\frac{h}{2}\right)^{(m+n+1)} \frac{[1+(-1)^{m+n}]}{(m+n+1)} \Bigg\}; \quad (n = 0, 1, 2), \end{aligned} \tag{5.13}$$

where the series on the right is convergent and

$$I^{\beta\nu}_{\eta\rho} = \delta^\beta_\eta b^\nu_\rho + b^\beta_\eta \delta^\nu_\rho - 2H\delta^\beta_\eta \delta^\nu_\rho. \tag{5.14}$$

We now proceed to obtain a first approximation to the constitutive equations (5.10) by retaining only the leading term on the right-hand side of (5.13) and introducing this result into the coefficients ${}_nD_{\alpha\beta\gamma\delta}$. Observing that the neglect of the series $\sum_{m=1}^{\infty}$ on the right-hand side of (5.13), in view of (5.12) and (5.14), implies the approximation $(h/R) \ll 1$ and that for $n = 1$ the first term of (5.13) vanishes identically, we have corresponding to the leading term in (5.13) the coefficients

$$\frac{1}{h}\,{}_0B^{\mu\beta\lambda\eta} = \frac{12}{h^3}\,{}_2B^{\mu\beta\lambda\eta} = \bar{A}^{\mu\beta\lambda\eta}, \quad {}_1B^{\mu\beta\lambda\eta} = 0. \tag{5.15}$$

* From the point of view of shell theory, this is indeed a weak restriction. For details and consequences of this restriction on the space of normal co-ordinates, see [1; Sec. 3].

If we now insert (5.15) into (5.8) and note that terms such as $b^{\eta}_{\beta}\,{}_2B^{\lambda\beta\rho\varepsilon}$ being of the same type as those arising from ${}_1B^{\mu\beta\lambda\eta}$ with $m = 1$, namely $(h^3/12)\bar{A}^{\mu\rho\lambda\delta}I^{\beta\eta}_{\rho\delta}$, are already omitted in obtaining (5.15), it then follows that corresponding to the leading term on the right-hand side of (5.13), the coefficients ${}_nD^{\lambda\eta\rho\varepsilon}$ read

$$\frac{1}{h}\,{}_0D^{\lambda\eta\rho\varepsilon} = \frac{12}{h^3}\,{}_2D^{\lambda\eta\rho\varepsilon} = \bar{A}^{\lambda\eta\rho\varepsilon}, \quad {}_1D^{\lambda\eta\rho\varepsilon} = 0 \tag{5.16}$$

and satisfy the symmetry relations $(5.8)_1$.

Thus, in view of (5.10) and (5.16),

$$\tilde{N}^{\alpha\beta} = h\bar{A}^{\alpha\beta\rho\varepsilon}\gamma_{(\rho\varepsilon)}, \quad M^{(\alpha\beta)} = \frac{h^3}{12}\bar{A}^{\alpha\beta\rho\varepsilon}\tilde{\kappa}_{\rho\varepsilon} \tag{5.17}$$

represent the first approximation to the constitutive equations (5.10) in the sense that only the leading term has been retained on the right-hand side of (5.13). The approximate constitutive equations (5.17), although involving different variables, have the same structural simplicity as those commonly referred to as Love's first approximation. However, unlike those of Love's first approximation, they satisfy all invariance requirements stated above and in addition, meet the reciprocity and variational theorems and all other general theorems which can be shown to hold when (5.10) is used.

It is of special significance that in spite of the manner in which the strain measures (3.1) and (3.2) and the resultants (3.5) and (3.6) are introduced, the boundary-value problems of the linear elastic shell theory, defined by means of the field equations and the boundary conditions derived here, involve only symmetric tensors. Moreover, the field equations consisting of (3.4), (3.8) and (5.10) or (5.17), as well as the natural boundary conditions (4.20) whenever these are relevant, are free of the anti-symmetric parts of the middle surface strains and the couple resultants. The indeterminacy of $\gamma_{[\alpha\beta]}$ and $M^{[\alpha\beta]}$ is of no concern here and is reminiscent of similar features in the theory of elasticity with couple stresses where, e.g. the anti-symmetric part of the stress tensor can be determined to within the scalar part of the couple stress.

6. COMPARISON WITH PREVIOUS RESULTS

We conclude the present paper by making a comparison with previous derivations [3, 4] which are aimed at a satisfactory first approximation theory. Since Sanders' work [3] is carried out in lines of curvature co-ordinates and since Koiter's results [4] in these co-ordinates are the same as Sanders [3]*, the comparison that follows mainly concerns Koiter's work.

The modified differential equations of equilibrium in the form (3.8) have previously been given by Sanders [3] in lines of curvature co-ordinates and in a more general form by both Sanders [17] and Leonard [18] who, however, employ different definitions for one of their new variables. Whereas they write $\tilde{N}^{\alpha\beta}$ in exactly the same manner as the equivalent of (3.5), namely $\tilde{N}^{\alpha\beta} = N^{\alpha\beta} - b^{\alpha}_{\lambda}M^{\lambda\beta}$, instead of (3.6) they define their second variable by $\tilde{M}^{\alpha\beta} = \frac{1}{2}(M^{\alpha\beta} + M^{\beta\alpha})$. In this connection, it should be noted that with the limitation to lines of curvature co-ordinates, equilibrium equations analogous to (3.8) are also given by Novozhilov [2] who employs the physical components of $\tilde{N}^{12} = \tilde{N}^{21}$, but instead of those of $\tilde{N}^{11}$ and $\tilde{N}^{22}$ he adopts the physical components of N^{11} and N^{22}.

* Although Koiter's and Sanders' results are the same in lines of curvature co-ordinates, their methods of derivation are not the same. Koiter's analysis, based on considerations of strain energy, is somewhat similar to Novozhilov's [2]. The approximate constitutive equations of Koiter [4] and Sanders [3] fulfill all invariance requirements while Novozhilov's constitutive relations [2] do not meet co-ordinate invariance.

Koiter's [4] definitions of stress and couple resultants have the form

$$n^{\alpha\beta} = N^{\alpha\beta} + \tfrac{1}{2}(b^{\beta}_{\gamma} M^{\gamma\alpha} - b^{\alpha}_{\gamma} M^{\gamma\beta}), \quad m^{\alpha\beta} = m^{\beta\alpha}. \tag{6.1}$$

It is clear that Sanders' couple resultants $\tilde{M}^{\alpha\beta}$ and $(6.1)_2$, assumed by Koiter [4], are identical to $M^{(\alpha\beta)}$ in (3.7). However, the structure of $(6.1)_1$ appears to be different from that of (3.5) which has also been used by both Sanders [17] and Leonard [18]. To reconcile between these, we first observe that $(2.7)_3$ is equivalent to

$$N^{\alpha\beta} - N^{\beta\alpha} = b^{\alpha}_{\lambda} M^{\lambda\beta} - b^{\beta}_{\lambda} M^{\lambda\alpha}. \tag{6.2}$$

Now, if we decompose $N^{\alpha\beta}$ into its unique symmetric and anti-symmetric parts, i.e.

$$N^{\alpha\beta} = N^{(\alpha\beta)} + N^{[\alpha\beta]}, \tag{6.3}$$

then, with the aid of (6.2), we arrive at

$$\begin{aligned} N^{(\alpha\beta)} &= \tfrac{1}{2}(N^{\alpha\beta} + N^{\beta\alpha}) = N^{\alpha\beta} + \tfrac{1}{2}(b^{\beta}_{\lambda} M^{\lambda\alpha} - b^{\alpha}_{\lambda} M^{\lambda\beta}), \\ N^{[\alpha\beta]} &= \tfrac{1}{2}(N^{\alpha\beta} - N^{\beta\alpha}) = \tfrac{1}{2}(b^{\alpha}_{\lambda} M^{\lambda\beta} - b^{\beta}_{\lambda} M^{\lambda\alpha}), \end{aligned} \tag{6.4}$$

the right-hand side of the first of which is identical with Koiter's $n^{\alpha\beta}$. For further comparison, consider the difference of (3.5) and $(6.4)_1$ which after simplification reads

$$\tilde{N}^{\alpha\beta} - N^{(\alpha\beta)} = \tfrac{1}{2} b^{\beta}_{\gamma} M^{\gamma\alpha} + \tfrac{1}{2} b^{\alpha}_{\gamma} M^{\gamma\beta} \tag{6.5}$$

and represents the difference between the symmetric $\tilde{N}^{\alpha\beta}$ and the unique symmetric part of $N^{\alpha\beta}$.

Koiter's strain measures

$$\rho_{\alpha\beta} = \tfrac{1}{2}(\kappa_{\alpha\beta} + \kappa_{\beta\alpha} - b^{\lambda}_{\alpha} \omega_{\beta\lambda} - b^{\lambda}_{\beta} \omega_{\alpha\lambda}), \quad \mathring{\gamma}_{\alpha\beta} = \mathring{\gamma}_{\beta\alpha}, \tag{6.6}$$

where $\omega_{\alpha\lambda} = \tfrac{1}{2}(v_{\alpha\|\lambda} - v_{\lambda\|\alpha}) \equiv \gamma_{[\alpha\lambda]}$, have similar structures to his resultants defined by (6.1). It follows from Koiter's work [4] that $\mathring{\gamma}_{\alpha\beta}$ in (6.6) is the same as $\gamma_{(\alpha\beta)}$ in (3.3) and to establish the connection between $\rho_{\alpha\beta}$ and $\tilde{\kappa}_{\alpha\beta}$, observe that the compatibility equation $(2.4)_3$ is equivalent to

$$\kappa_{\alpha\beta} - \kappa_{\beta\alpha} = b^{\nu}_{\beta} \gamma_{\nu\alpha} - b^{\nu}_{\alpha} \gamma_{\nu\beta}, \tag{6.7}$$

decompose $\kappa_{\alpha\beta}$ as in (3.10), and with the aid of (6.7) write

$$\begin{aligned} \kappa_{(\alpha\beta)} &= \tfrac{1}{2}(\kappa_{\alpha\beta} + \kappa_{\beta\alpha}) = \kappa_{\alpha\beta} + \tfrac{1}{2}(b^{\nu}_{\alpha} \gamma_{\nu\beta} - b^{\nu}_{\beta} \gamma_{\nu\alpha}), \\ \kappa_{[\alpha\beta]} &= \tfrac{1}{2}(\kappa_{\alpha\beta} - \kappa_{\beta\alpha}) = \tfrac{1}{2}(b^{\nu}_{\beta} \gamma_{\nu\alpha} - b^{\nu}_{\alpha} \gamma_{\nu\beta}). \end{aligned} \tag{6.8}$$

Since $(6.6)_1$ may be also written as

$$\begin{aligned} \rho_{\alpha\beta} &= \tfrac{1}{2}(\kappa_{\alpha\beta} + \kappa_{\beta\alpha}) - \tfrac{1}{2}\{\tfrac{1}{2}(2 b^{\lambda}_{\beta} \gamma_{[\alpha\lambda]} - 2 b^{\lambda}_{\alpha} \gamma_{[\lambda\beta]})\} \\ &= \kappa_{(\alpha\beta)} + \tfrac{1}{2}(b^{\lambda}_{\alpha} \gamma_{[\lambda\beta]} + b^{\lambda}_{\beta} \gamma_{[\lambda\alpha]}), \end{aligned} \tag{6.9}$$

with the aid of (6.8), we have

$$\begin{aligned} \tilde{\kappa}_{\alpha\beta} - \kappa_{(\alpha\beta)} &= -\tfrac{1}{2}(b^{\nu}_{\alpha} \gamma_{\nu\beta} + b^{\nu}_{\beta} \gamma_{\nu\alpha}), \\ \tilde{\kappa}_{\alpha\beta} - \rho_{\alpha\beta} &= -\tfrac{1}{2}[b^{\lambda}_{\alpha} \gamma_{(\lambda\beta)} + b^{\lambda}_{\beta} \gamma_{(\lambda\alpha)}], \\ \kappa_{(\alpha\beta)} - \rho_{\alpha\beta} &= \tfrac{1}{2}(b^{\lambda}_{\alpha} \gamma_{[\lambda\beta]} + b^{\lambda}_{\beta} \gamma_{[\lambda\alpha]}). \end{aligned} \tag{6.10}$$

To complete the desired comparison, we note that in deriving his uncoupled approximate constitutive relations [similar to (5.17)] on the basis of strain energy considerations, Koiter [4] argues that terms of the type $b^{\lambda}\overset{\circ}{\gamma}_{\lambda\beta}$ are unimportant when the strain energy function is approximated to one without coupling between the strain measures. Koiter further suggests in [4] that expressions for the change of curvature can often be simplified by addition or subtraction of terms of the type $b^{\lambda}_{\alpha}\overset{\circ}{\gamma}_{\lambda\beta}$ which is equivalent to $b^{\lambda}_{\alpha}\gamma_{(\lambda\beta)}$. It should now be evident that if we accept Koiter's argument, then the differences between Koiter's constitutive relations and (5.17), in view of $(6.4)_1$ and $(6.10)_2$, are indeed reconcilable.

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Note added in press—After the paper was in press, the author's attention was called to a recent paper by B. Budiansky and J. L. Sanders, Jr. [*Progress in Applied Mechanics, The Prager Anniversary Volume*, p.129, Macmillan Co. (1963)] which deals with a first-order linear shell theory. Although the two papers employ fundamentally different approaches, they partly overlap in scope and some of the results are the same.

Résumé—La mémoire concerne une nouvelle façon de déduire les équations générales de la théorie linéaire des coques élastiques, d'après l'hypothèse de Kirchhoff–Love. L'ensemble du problème de la théorie des coques est refondu et mis en fonction de nouvelles variables, aussi bien pour les déformations mesurées que pour les résultantes des efforts et des couples. La déduction exacte des équations constitutives et de leur première approximation, laquelle satisfait a toutes les exigences de l'invariance, a fait l'objet d'une attention particulière. Les conditions aux limites naturelles pour les résultantes des efforts et des couples et l'ensemble des équations de compatibilité, d'équilibre et constitutives (ou leur première approximation) ne contiennent que des tenseurs symétriques; elles sont d'ailleurs remarquables par leur manque d'éléments antisymétriques dans l'expression de la déformation moyenne de la surface et des résultantes des couples.

Zusammenfassung—Diese Abhandlung befasst sich mit einer neuen Ableitung der allgemeinen Gleichungen der Lineartheorie von elastischen Schalen nach der Kirchhoff–Love'schen Hypothese. Das gesamte Randwertproblem der Schalentheorie wird vermittels neuer Variablen für die Dehnungsmasse sowie für die Spannungs- und Kräftepaarresultanten neu gestellt. Besonderer Wert wird einer genauen Ableitung der darstellenden Gleichungen und ihrer ersten Annäherung—die allen Invarianzbedürfnissen entsprechen—beigemessen. Die natürlichen Randbedingungen für die Spannungs- und Kräftepaarresultanten sowie für alle aus Verträglichkeit, Gleichgewicht und den darstellenden Gleichungen (oder ihren ersten Annäherungen) bestehenden Feldgleichungen haben nur mit symmetrischen Tensoren zu tun und sind ausserdem in bemerkenswerter Weise frei von den antisymmetrischen Teilen der Dehnungen der mittleren Oberfläche sowie der Kräftepaarresultanten.

Sommario—La presente pubblicazione si occupa di una nuova derivazione delle equazioni generali della teoria lineare dei gusci elastici in base all'ipotesi Kirchhoff–Love. Tutto il problema del valore limite della teoria dei gusci viene riproposto in termini di nuove variabili per le misurazioni delle sollecitzioni oltre che delle deformazioni e delle risultanti di coppia. Si presta particolare attenzione ad una esatta derivazione delle equazioni costitutive e della loro prima approssimazione che soddisfa tutte le esigenze di invarianza. Le condizioni limite naturali per la deformazione e le risultanti di coppia e tutte le equazioni di campo consistenti di compatibilità, equilibrio ed equazioni costitutive (o le loro prime approssimazioni) comportano solo tensori simmetrici e sono, per di più, notevolmente esenti delle parti antisimmetriche sia delle sollecitazioni della superficie media che delle risultanti di coppia.

Абстракт — Рассматривается новый вывод общих уравнений линейной теории эластичных оболочек с точки зрения гипотезы Киркгофа–Лова.

В целом проблема значения границ в теории оболочек переведена на язык новых переменных, измеряющих деформации, напряжение и результанты пары. Особое внимание обращено на точный вывод уравнений строения и их первых приближений, встречающих все требования инвариантности. Натуральные условия границ для напряжения и результантов пары также как все уравнения поля, состоящие из уравнений согласования, равновесия и состава (или их первых приближений) включают только симметричные тензоры и кроме того они являются замечательно независимыми от анти-симметричных частей как средне-поверхностных деформаций так и результантов пар.